CSE-476: Introduction to Natural Language Processing
Quiz 2

Multiple Choice Questions (Single correct)

Instructions: Select the correct option with a short explanation.

Q1. What is the main difference between generative and discriminative classifiers?

- (A) Generative models model $P(x, y)$, while discriminative models model $P(y \mid x)$
- (B) Discriminative models learn the data distribution and use that knowledge for classification
- (C) Discriminative models require no training data
- (D) Generative models are less explainable

Q2. Which technique is used to mitigate overfitting by penalizing large weights?

- (A) Laplace smoothing
- (B) Regularization
- (C) TF-IDF
- (D) Softmax normalization

Q3. What is the limitation of the perceptron algorithm?

- (A) It cannot use vector representations
- (B) It only works for linearly separable data
- (C) It requires probabilistic outputs
- (D) It cannot be trained with gradient descent

Q4. Why is Laplace smoothing important?

- (A) It improves cosine similarity
- (B) It reduces dimensionality
- (C) It avoids zero probabilities for unseen words
- (D) It speeds up gradient descent

Q5. What happens in stochastic gradient descent (SGD) compared to gradient descent (GD)?

(A) SGD updates weights using the entire dataset every step
(B) SGD updates weights using one or a subset of training examples
(C) GD converges faster because updates are noisier
(D) SGD cannot be used with neural networks

Q6. What is the core intuition behind word embeddings?

(A) Words are represented by their dictionary definitions
(B) Words that appear in similar contexts have similar meanings
(C) Words are clustered by part of speech
(D) Words are ranked by frequency

Q7. What is the primary purpose of the Softmax function?

(A) To penalize large model weights
(B) To reduce feature dimensionality
(C) To select the class with the largest raw score without normalization
(D) To convert raw class scores into a valid probability distribution

Q8. In the perceptron algorithm, when are the weights updated?

(A) After every training example
(B) Only when the prediction is correct
(C) Only when the prediction is incorrect
(D) Only after one full pass over the data

Q9. What is the main goal of TF-IDF weighting?

(A) Increase the weight of very common words
(B) Represent word order
(C) Convert sparse vectors into dense embeddings
(D) Highlight words that are frequent in a document but rare across documents

Q10. Why is cosine similarity preferred over dot product when comparing word vectors?

(A) It ignores vector length while calculating similarity
(B) It increases the influence of high-frequency words
(C) It converts similarity scores into probabilities
(D) It reduces the dimensionality of word embeddings

End of Paper